

The Sharp Free Approximation Rate for $(1 - x)(1 - y)$

An exact recursion and a full candidate answer to the example in arXiv:2209.10373

Automated mathematics run `fa_banach_001`, lane 5

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Status. Candidate full solution to the explicit sharpness question for $f(x, y) = (1 - x)(1 - y)$, likely valid and awaiting human review. The theorem below proves the stronger asymptotic $\sqrt{n} c_n \rightarrow 1$.

1 Source question

The source is P. Arora, M. Augat, M. T. Jury, and M. Sargent, *An Optimal Approximation Problem for Free Polynomials*, arXiv:2209.10373, published in *Proceedings of the American Mathematical Society* **152** (2024), 455–470 [1]. Free words are declared orthonormal, and the authors consider

$$c_n = \min_{\deg p \leq n} \|pf - 1\|_2^2.$$

In Section 3.1 (arXiv PDF page 14), they observe that for $f(x, y) = (1 - x)(1 - y)$ one can obtain $c_n \lesssim n^{-1/2}$, but state that they do not know whether the exponent $1/2$ is sharp. The relevant source passage is reproduced below.

3.1. Remarks and Questions. In general, the exponent $p = \frac{1}{3\ell-1}$ appearing in Theorem 1.1 (where ℓ is the number of irreducible factors of F) is not sharp. Already in one variable, the bound $c_n \lesssim \frac{1}{n}$ holds regardless of the number of factors, as was already mentioned in the introduction.

It is possible to run the proof presented here in the one-variable case, making more careful choices of the polynomials constructed in the induction step, to recover the uniform $\frac{1}{n}$ bound in that case, but this relies crucially on the commutativity of the polynomial ring $\mathbb{C}[z]$.

At the moment, we do not know if the same is true in the free setting, or even if there is a uniform power p so that $\|P_n F - I\|_2^2 \lesssim \frac{1}{n^p}$ independently of F . Also, observe that our proof does not actually construct the optimal approximants for any given f , we only construct a sequence p_n obeying the claimed bounds, so that the optimal approximants must be at least as good. We have computed the optimal approximants (with computer assistance) in some simple examples, but the results do not appear enlightening and we have not included them.

The sharp value of p seems difficult to calculate even for relatively simple cases. For example, for the free polynomial in two variables $f(x, y) = (1 - x)(1 - y)$ one can construct polynomials p_n (by essentially the method used in the induction step of the proof, but making more careful choices of σ_n, r_n) for which

$$\|p_n(x, y)(1 - x)(1 - y) - 1\|_2^2 \lesssim \frac{1}{n^{1/2}},$$

which improves on the bound $\frac{1}{n^{1/3}}$ provided by the theorem, but we do not know if this value $p = 1/2$ is sharp.

Figure 1: The sharpness question in Section 3.1 of the source, arXiv PDF page 14.

2 Definitions and statement

Let $\mathbb{F}_2^+$ be the free monoid on the letters x, y , with empty word $\emptyset$. Write e_w for the word basis of $\ell^2(\mathbb{F}_2^+)$, and let

$$\mathcal{H}_n = \text{span}\{e_w : |w| \leq n\}.$$

A free polynomial is identified with its coefficient vector in this Hilbert space. The optimal squared error for the source example is

$$c_n := \min_{p \in \mathcal{H}_n} \|p(1-x)(1-y) - e_\emptyset\|_2^2.$$

Theorem 1. *Define two positive sequences $(e_m)_{m \geq -1}$ and $(\kappa_m)_{m \geq -1}$ by*

$$e_{-1} = 1, \quad \kappa_{-1} = \frac{1}{2},$$

and, for $m \geq 0$,

$$\kappa_m = \kappa_{m-1} + \frac{e_{m-1}}{1 + e_{m-1}}, \tag{1}$$

$$e_m = 1 + \frac{\kappa_{m-1}e_{m-1}}{\kappa_{m-1} + e_{m-1}}. \tag{2}$$

Then the optimal errors for $f(x, y) = (1-x)(1-y)$ satisfy the exact formula

$$\boxed{c_n = \frac{1}{e_n}} \quad (n \geq 0).$$

Moreover,

$$\boxed{\lim_{n \rightarrow \infty} \sqrt{n} c_n = 1}.$$

Consequently the exponent $p = 1/2$ in the source question is sharp, and its leading constant is one.

For orientation, the first exact values are

$$c_0 = \frac{3}{4}, \quad c_1 = \frac{7}{11}, \quad c_2 = \frac{14}{25}, \quad c_3 = \frac{56}{111}.$$

3 Proof intuition

Right multiplication by $(1-x)(1-y)$ sends the coefficient at a word w to a four-point pattern supported at w, wx, wy, wxy . Orthogonality to all such patterns rooted at depth at most n is therefore the local relation

$$h_w - h_{wx} - h_{wy} + h_{wxy} = 0.$$

Projection duality says that c_n^{-1} is the least squared norm of a vector satisfying all these relations and normalized by $h_\emptyset = 1$.

This is an electrical-energy-style minimization on a finite binary tree. A rooted subtree has only two boundary data relevant to its parent: its root and its y -child. Eliminating all other values turns the entire tree into a 2×2 positive quadratic form. Its Schur complement is summarized by one scalar κ_m , while the unconstrained-root energy is e_m . This gives the exact recursion in Theorem 1.

For the asymptotics, $e \mapsto 1 + \kappa e / (\kappa + e)$ has a moving fixed point $r(\kappa)$ defined by $r^2 - r = \kappa$. The recursion keeps e_m below this fixed point, while any fixed relative gap is closed by a uniformly positive amount at each step; the fixed point itself moves only by $O(1/r)$. Hence $e_m / r(\kappa_m) \rightarrow 1$. Since $\kappa_m \sim m$, the claimed limit follows.

4 Exact dual recursion

Let $R_x e_w = e_{wx}$ and $R_y e_w = e_{wy}$ be the right creation operators, and put

$$A = (I - R_y)(I - R_x).$$

Thus $Ap = p(1 - x)(1 - y)$ and

$$c_n = \text{dist}(e_\emptyset, A\mathcal{H}_n)^2.$$

Let $L_n = (A\mathcal{H}_n)^\perp$. Standard Hilbert-space projection duality gives

$$\frac{1}{c_n} = \min\{\|h\|_2^2 : h \in L_n, h_\emptyset = 1\}. \quad (3)$$

Indeed, if g is the orthogonal projection of $e_\emptyset$ onto L_n , then $c_n = \|g\|^2$; among $h \in L_n$ with $\langle e_\emptyset, h \rangle = 1$, equality in Cauchy-Schwarz is attained by $h = g/\|g\|^2$.

For every word w ,

$$(A^*h)_w = h_w - h_{wx} - h_{wy} + h_{wxy}.$$

Consequently $h \in L_n$ precisely when

$$h_w - h_{wx} - h_{wy} + h_{wxy} = 0 \quad (|w| \leq n). \quad (4)$$

Only coordinates of depth at most $n + 2$ occur in these constraints. Setting all deeper coordinates to zero preserves feasibility and decreases the norm, so the minimum in (3) is a finite-tree problem.

We now solve that problem. For $m \geq -1$, let $E_m(a)$ be the minimum energy in a rooted binary subtree whose root has value a , with (4) imposed through relative depth m . Let $F_m(a, z)$ be the same minimum when the value z at the root's y -child is also fixed. The case $m = -1$ means that no relation is imposed, so

$$E_{-1}(a) = |a|^2, \quad F_{-1}(a, z) = |a|^2 + |z|^2.$$

Tree symmetry and homogeneity give

$$E_m(a) = e_m |a|^2, \quad F_m(a, z) = \begin{bmatrix} \bar{a} & \bar{z} \end{bmatrix} Q_m \begin{bmatrix} a \\ z \end{bmatrix},$$

where Q_m is a real symmetric positive matrix. The data are real, so this Hermitian formulation also shows that allowing complex coefficients does not change the scalar recursion.

Suppose

$$Q_{m-1} = \begin{bmatrix} A & B \\ B & C \end{bmatrix}, \quad \kappa_{m-1} := \frac{\det Q_{m-1}}{A + 2B + C}.$$

At a root with value a , write b, c, d for the values at wx, wy, wxy . The root relation is $d = b + c - a$. If the y -child is fixed at $z = c$, the two child subtrees are disjoint and therefore

$$F_m(a, z) = |a|^2 + E_{m-1}(z) + \min_b F_{m-1}(b, b + z - a). \quad (5)$$

For any $s \in \mathbb{C}$, direct completion of the square gives

$$\min_b F_{m-1}(b, b + s) = \frac{AC - B^2}{A + 2B + C} |s|^2 = \kappa_{m-1} |s|^2.$$

Hence

$$Q_m = \begin{bmatrix} 1 + \kappa_{m-1} & -\kappa_{m-1} \\ -\kappa_{m-1} & e_{m-1} + \kappa_{m-1} \end{bmatrix}. \quad (6)$$

If the y -child is not fixed, a second one-variable minimization in (5) yields

$$E_m(a) = |a|^2 + \min_z (e_{m-1}|z|^2 + \kappa_{m-1}|z - a|^2) = \left(1 + \frac{\kappa_{m-1}e_{m-1}}{\kappa_{m-1} + e_{m-1}}\right) |a|^2.$$

This is (2). From (6),

$$\frac{\det Q_m}{(Q_m)_{11} + 2(Q_m)_{12} + (Q_m)_{22}} = \kappa_{m-1} + \frac{e_{m-1}}{1 + e_{m-1}},$$

which is (1). Since $Q_{-1} = I$, its initial Schur-complement parameter is $\kappa_{-1} = 1/2$.

Finally, the minimum in (3) is exactly the root energy $E_n(1) = e_n$. Therefore $c_n = 1/e_n$, proving the exact part of Theorem 1.

5 Sharp asymptotics

We prove that $e_n \sim \sqrt{n}$. Write $k_n = \kappa_n$ for brevity and define

$$r(k) = \frac{1 + \sqrt{1 + 4k}}{2}, \quad r(k)^2 - r(k) = k.$$

For fixed $k > 0$, set

$$\Phi_k(u) = 1 + \frac{ku}{k + u}.$$

The number $r(k)$ is the positive fixed point of Φ_k , and Φ_k is increasing. Since $e_{-1} = 1 < r(k_{-1})$ and $k_n > k_{n-1}$, induction gives

$$1 \leq e_n \leq r(k_n) \quad (n \geq -1). \quad (7)$$

Indeed, $e \leq r(k)$ implies $\Phi_k(e) \leq r(k) < r(k')$ for every $k' > k$. It also implies monotonicity, because

$$\Phi_k(e) - e = \frac{k + e - e^2}{k + e} \geq 0. \quad (8)$$

The recursion for k_n , together with $e_n \geq 1$, gives

$$\frac{n+2}{2} \leq k_n < n + \frac{3}{2} \quad (n \geq 0). \quad (9)$$

We also record a useful elementary lower bound. Since $k_0 = 1$ and $e_0 = 4/3$, induction shows

$$e_n \geq \frac{1}{2}\sqrt{k_n} \quad (n \geq 0). \quad (10)$$

To verify the induction step, suppose $k \geq 1$ and $e \geq \frac{1}{2}\sqrt{k}$, and put $k' = k + e/(1 + e) \leq k + 1$. If $e \geq \frac{1}{2}\sqrt{k+1}$ there is nothing to prove, by (8). Otherwise $e^2 < (k+1)/4$ and $e < \frac{1}{2}\sqrt{k+1} \leq k$, so

$$\Phi_k(e) - e = \frac{k + e - e^2}{k + e} > \frac{k - (k+1)/4}{2k} \geq \frac{1}{4}.$$

But

$$\frac{1}{2}(\sqrt{k+1} - \sqrt{k}) = \frac{1}{2(\sqrt{k+1} + \sqrt{k})} < \frac{1}{4},$$

which proves $\Phi_k(e) \geq \frac{1}{2}\sqrt{k+1} \geq \frac{1}{2}\sqrt{k'}$.

Equations (7)–(10) already give explicit two-sided estimates

$$\frac{1}{1 + \sqrt{n + 3/2}} \leq c_n \leq \frac{2\sqrt{2}}{\sqrt{n + 2}}. \quad (11)$$

Thus the exponent $1/2$ is sharp even without identifying the leading constant. We finish by proving that the constant is one.

The lower bounds above imply $e_n \rightarrow \infty$. Hence

$$k_n = \frac{1}{2} + \sum_{j=-1}^{n-1} \frac{e_j}{1 + e_j} \implies \frac{k_n}{n} \longrightarrow 1$$

by Cesàro convergence. It remains to compare e_n with $r(k_n)$.

Fix $0 < \varepsilon < 1$ and set $s = (1 - \varepsilon)r$, where $r = r(k)$. The increment $\Phi_k(u) - u$ is decreasing in $u > 0$, and at s it equals

$$\Phi_k(s) - s = \frac{(r - s)(r + s - 1)}{k + s} = \varepsilon \frac{(2 - \varepsilon)r - 1}{r - \varepsilon} \longrightarrow \varepsilon(2 - \varepsilon) > 0. \quad (12)$$

On the other hand, since $k_{n+1} - k_n < 1$ and $r'(k) = 1/(2r(k) - 1)$,

$$r(k_{n+1}) - r(k_n) \leq \frac{1}{2r(k_n) - 1} \longrightarrow 0. \quad (13)$$

Choose $d > 0$ and N so that, for $n \geq N$, the boundary increment in (12) is at least d , while the increase of $(1 - \varepsilon)r(k_n)$ is less than $d/2$. Whenever $e_n \leq (1 - \varepsilon)r(k_n)$, its next increment is at least d , so its gap below that moving boundary decreases by at least $d/2$. It crosses the boundary after finitely many steps. Once it has crossed, monotonicity of Φ_{k_n} and the same two estimates show that it remains above the boundary. Therefore

$$\liminf_{n \rightarrow \infty} \frac{e_n}{r(k_n)} \geq 1 - \varepsilon.$$

Since (7) gives the reverse upper bound and ε is arbitrary,

$$\frac{e_n}{r(k_n)} \longrightarrow 1.$$

Now $r(k_n)/\sqrt{k_n} \rightarrow 1$ and $k_n/n \rightarrow 1$, so $e_n/\sqrt{n} \rightarrow 1$. The identity $c_n = 1/e_n$ completes the proof of Theorem 1.

6 Verification notes

The exact recursion was checked independently against the original primal least-squares problem. The script `code/verify_recurrence.py` builds the four-term coefficient column of $w(1 - x)(1 - y)$ for every $|w| \leq n$, forms its Gram matrix over the rational numbers, solves the normal equations, and compares the resulting residual with $1/e_n$. The values agree exactly for every tested degree. The script also iterates the scalar recurrence in floating point to display convergence of $\sqrt{n}/e_n$ to one. These checks are corroborative only; no numerical assertion is used in the proof.

The most delicate formal points are local and exposed in the packet: the right-multiplication order in the definition of A , the fact that wxy is the y -child of the x -child in the tree split, and the Schur-complement coefficient in the one-variable minimization. The separate `verification.md` records these checks and the tested range.

7 Scope and novelty check

This result fully answers the source paper’s explicit question whether the exponent $1/2$ is sharp for $(1 - x)(1 - y)$. It does not answer the broader question in the same section asking whether there is a uniform positive exponent valid independently of the free polynomial F .

The bounded novelty search was performed on 9 August 2026. The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2209.10373, the source title, “optimal polynomial approximants”, “free polynomial”, and the explicit factor $(1 - x)(1 - y)$. Web/arXiv searches used the exact title and example, the author names, the journal DOI 10.1090/proc/16474, and combinations with “sharp”, “rate”, and “one-half”. The available citation trail surfaced later work on classical Hardy-space optimal approximants that cites the source as background, but no paper treating this free example or giving the recursion above. The search is bounded rather than exhaustive, so bibliographic novelty remains subject to specialist review.

8 Human-review recommendation

Review as a full solution to the concrete source example. First check (3) and the coordinate identity for A^* . Then check the subtree split (5), particularly that the fixed value $b + z - a$ is the y -child of the x -child. Finally verify the moving-boundary argument after (12). If those points pass, the exact recurrence and the asymptotic constant follow without any external mathematical input.

References

- [1] P. Arora, M. Augat, M. T. Jury, and M. Sargent, *An Optimal Approximation Problem for Free Polynomials*, Proc. Amer. Math. Soc. **152** (2024), no. 2, 455–470; arXiv:2209.10373, doi:10.1090/proc/16474.